Mapping 0il price risk, for a portfolio management strategy

A Structural Sector-Rotation Overlay for Indian Equities

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Abstract

An aggregate move in the price of crude oil bundles together at least three economically distinct forces: a disruption to global supply, a swing in global demand, and a precautionary or speculative demand for oil itself. This study separates those forces using the structural vector autoregression of Kilian [7], measures how each one propagates into ten Indian equity and safe-haven sleeves through Jordà [2005] local projections, and converts the resulting exposure map into a monthly portfolio overlay. The overlay tilts an equal-weight sector book toward oil-resilient sectors and switches on a conditional gold and liquid-bond hedge when oil-shock intensity is high. Out of sample (March 2023 to February 2026) the Black-Litterman version of the overlay lifts the net Sharpe ratio from 1.27 to 1.85 and cuts the portfolio's oil-variance share by roughly half. The study is deliberate about what the evidence does and does not support: the structural variance reduction is tight and robust, while the value of *timing* the hedge cannot be statistically separated from a constant defensive tilt at this sample size. Both findings are reported in full.

1 Executive Summary

The problem. A single number for "the oil price" hides three different economic events. This study separates an oil-supply disruption, a global-demand swing, and an oil-specific demand shock, measures how each one hits ten Indian equity and safe-haven sleeves, and uses that map to run a monthly portfolio overlay that carries less oil risk.

What was built. Three portfolios run in parallel and rebalance monthly: an equal-weight benchmark (A), a transparent linear tilt (B-Lin) as a control, and the headline Black-Litterman overlay with a conditional gold and liquid-bond hedge (B-BL). All parameters were fixed before any out-of-sample result was examined.

Headline findings.

- **Better risk-adjusted return out of sample.** The BL overlay lifts the net Sharpe ratio from 1.27 to 1.85 (March 2023 to February 2026) and raises annual net return from 11.9% to 16.2%, at *lower* volatility (8.7% versus 9.4%).
- **Roughly half the oil risk.** The overlay cuts the portfolio's oil-variance share from 1.75% to 0.84%, a reduction with a tight bootstrap interval that excludes zero.
- **Cheap and conditional.** It does this at one-seventeenth the turnover of the linear control, and its gold weight tracks oil-shock intensity with a correlation of +0.68.
- **Robust where it matters.** The overlay improves the Sharpe ratio in all four walk-forward folds, retains 91 to 97% of its protection when target episodes are excised from estimation, and stays positive across the full cost and parameter grids.

What the evidence does not support. Two honest negatives bound the claims. The value of *timing* the hedge cannot be statistically separated from a constant defensive tilt at this sample size (about three independent oil episodes), and a supply-shock placebo protects nearly as well as the oil-specific signal. The structural variance reduction is solid; the timing alpha is unproven. The remainder of the report develops each of these points in full.

2 Motivation and Research Question

Oil is the single most scrutinised macro price in the world, yet most equity risk models treat "the oil price" as one number. That is a mistake. A barrel that rises because a pipeline was sabotaged is not the same event, for a portfolio, as a barrel that rises because global growth is accelerating. The first is a cost shock that squeezes margins in energy-intensive sectors; the second is a demand signal that often lifts cyclical equities even as oil climbs. Kilian [7] formalised this intuition for the crude market and showed that the three underlying shocks, supply disruption, aggregate demand,

and oil-specific demand, have opposite signs and different time profiles in their effect on macro aggregates and, as Kilian and Park [8] document, on the stock market.

This project asks a practical question. If those three channels can be identified in real time from global data, can an Indian equity investor use the decomposition to build a portfolio that carries less oil risk and survives oil-driven drawdowns better, without giving up return? The answer developed here is a qualified yes. The decomposition produces a stable and economically sensible map of sector exposures, and a portfolio overlay built on that map measurably reduces oil-variance exposure. The weaker part of the claim, that dynamically *timing* the hedge beats a static defensive posture, is treated as an open question rather than a result, because the data do not contain enough independent oil episodes to settle it.

The pipeline runs in three stages, each of which is described in its own section: a structural VAR that extracts the three shocks (Section 4), a local-projection step that maps shocks to sector responses (Section 5), and a portfolio construction layer that turns the exposure map into monthly weights (Section 6). Section 7 reports performance and Section 8 subjects every headline claim to a battery of robustness tests.

3 Data

The macro block follows Kilian's specification as closely as the public data allow. Global crude production comes from the U.S. Energy Information Administration, real global activity from the Dallas Fed Index of Global Real Economic Activity, the nominal oil price from the EIA refiner acquisition cost series, and the U.S. Consumer Price Index from FRED is used to deflate it. The three transformed inputs are the monthly log-growth of production, the activity index in levels, and the CPI-deflated log real oil price. After transformation and alignment the macro sample runs from February 1993 to February 2026, giving 397 monthly observations.

The equity block consists of ten monthly total-return series downloaded from Yahoo Finance: seven Nifty sector indices (IT, Bank, Energy, Pharma, Auto, Metal, FMCG), the broad Nifty 50, plus two safe-haven sleeves, a gold exchange-traded fund and a liquid overnight-cash fund that stands in for short-duration bonds.[1] The equity sample begins in October 2007, constrained by the youngest index, and realised volatility is computed for each sleeve from daily returns inside each month. Table 1 reports the median annualised volatility of each sleeve and makes the risk hierarchy clear: Metal is the most volatile sleeve and the liquid-cash fund is, by construction, almost riskless.

Table 1: Median annualised realised volatility by sleeve, 2007–2026.

Sleeve	Vol (%)	Sleeve	Vol (%)
Metal	24.8	Auto	17.8
Bank	19.1	Energy	17.6
IT	19.0	Pharma	16.1
FMCG	14.3	Gold	11.8
Nifty 50	14.3	LiquidBond	0.9

A limitation is acknowledged at the outset rather than buried: the index constituents are current, not point-in-time, so the equity series carry survivorship bias. This biases return levels upward by an unknown amount, most likely for the younger sleeves (Metal and Auto entered the index universe after 2011). The study's central claims are therefore framed around *risk reduction* and *relative* behaviour during stress episodes, quantities far less sensitive to survivorship than raw return levels.

4 Stage One: The Kilian Structural VAR

Figure 1 shows the three transformed inputs that feed the model. Production growth is noisy and mean-reverting, real activity moves in long cyclical swings, and the real oil price traces the familiar arc of the 2000s super-cycle, the 2014 collapse, the 2020 shock, and the 2022 spike. The structural VAR's job is to disentangle which of these co-movements reflect supply, demand, or oil-specific forces.

4.1 Identification

The first stage estimates a three-variable VAR in production growth, real activity, and the real oil price, and recovers structural shocks through a recursive (Cholesky) ordering. The ordering encodes three economic timing assumptions that Kilian [7] argued are defensible at the monthly frequency. Crude production is placed first, because physical extraction is slow to respond within a single month to news about demand or prices. Real activity is placed second, so it can react contemporaneously to supply but not to oil-specific demand. The real oil price is placed last, absorbing as a residual any price movement not explained by the

[1] A long-duration government-bond ETF was attempted but dropped: it only launched around 2015 and lacked the history needed for estimation. Gold and liquid cash therefore carry the entire defensive sleeve.

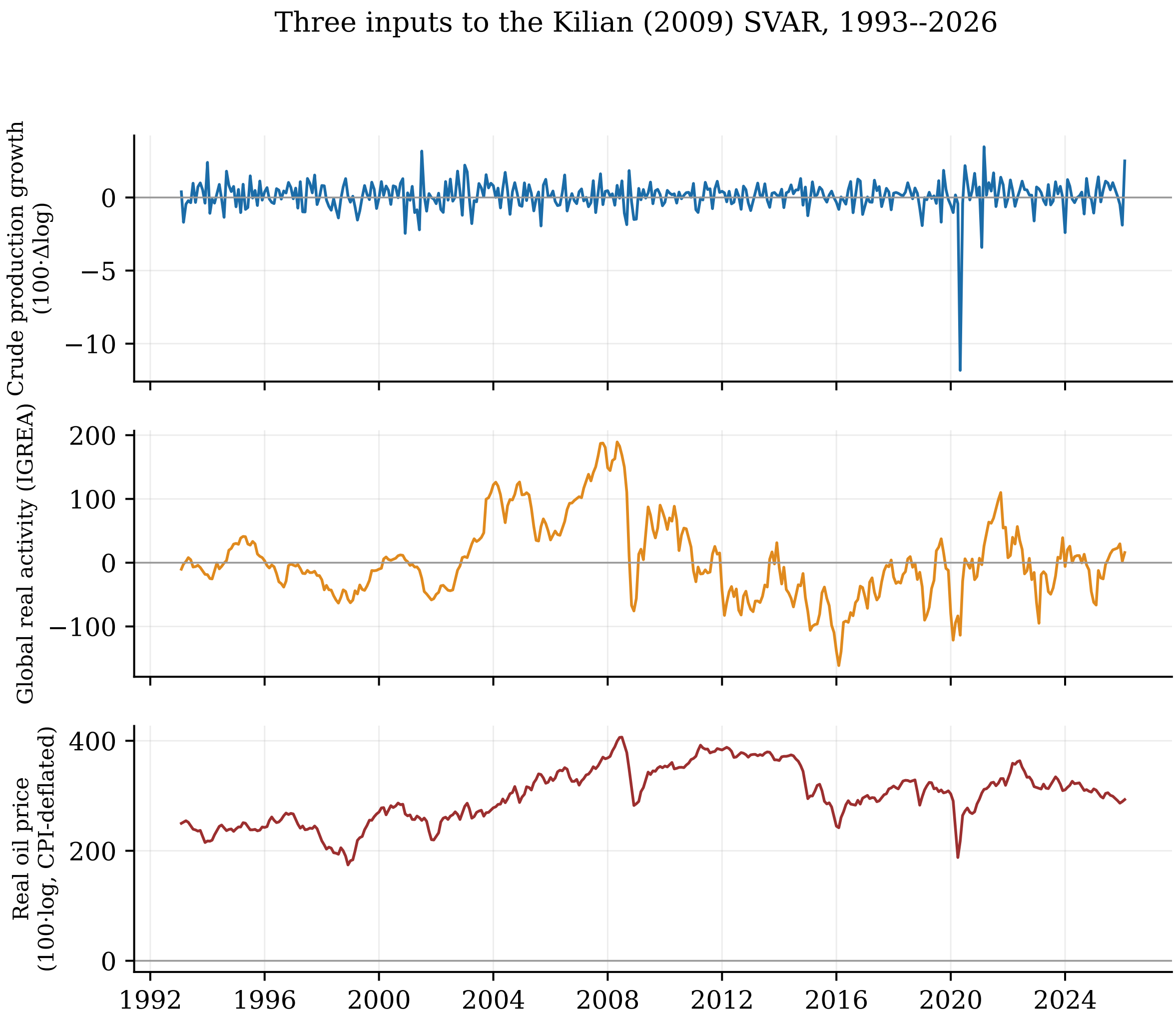

Figure 1: The three inputs to the SVAR over the full macro sample: crude production growth, the Dallas Fed global real activity index, and the CPI-deflated real oil price.

first two, which is precisely the oil-specific demand shock. Formally, with reduced-form residuals u_t and structural shocks ε_t, the mapping is

$$u_t = P\,\varepsilon_t, \qquad PP' = \Sigma_u, \qquad P \text{ lower triangular},$$

so that P is the Cholesky factor of the residual covariance matrix and the shocks are orthogonal by construction.

4.2 Estimation and lag choice

Contrary to Kilian's original 24-lag annual-frequency setup, the order here is chosen by information criterion on the actual sample. The Akaike criterion selects three lags and the model is stable (all companion-matrix eigenvalues inside the unit circle). Before estimation, standard pre-tests confirm that production growth and real activity are stationary; the real oil price fails an Augmented Dickey-Fuller test ($p = 0.16$) and is flagged by KPSS. Following Kilian [7] and Hamilton [5], the price is kept in levels rather than differenced: differencing would redefine the third shock as a shock to the *change* in price, a different economic object, and would discard the price-level information that identifies it. A differenced-price VAR is estimated separately as a specification check and produces qualitatively similar impulse responses, indicating the near-unit-root is not driving the results.

4.3 Outlier-robust dummy selection

A small number of months are so extreme that they distort the Cholesky factor for the entire sample. The clearest case is the spring of 2020, when an OPEC+ price war and a demand collapse coincided and production responded *within the month*, directly violating the recursive zero restriction. Rather than choose dummies by eye, the study detects them with a robust Mahalanobis distance based on the Minimum Covariance Determinant estimator [13], which fits the covariance on the most

Table 2: Impulse dummies selected by the outlier-robust convergence loop.

Month	Event	Round
2020-03	COVID demand collapse and price war	1
2020-04	Peak lockdown, demand down ~30 mb/d	1
2020-05	Negative WTI front-month episode	1
2008-10	Global financial crisis (Lehman)	2
2021-02	Texas deep freeze (Winter Storm Uri)	3
2023-03	SVB collapse and banking stress	4
2020-01	Early COVID demand fears	5

concentrated 87.5% of months so that the extreme observations cannot mask themselves. A sequential loop in the spirit of Lütkepohl [10] then adds one impulse dummy at a time and re-estimates until no residual exceeds the inclusion threshold. The procedure converges on seven dummies, listed in Table 2.

The dummies matter. Removing them changes the contemporaneous standard deviation of the supply shock by 19%, which is a change in the *unit* of the shock series carried into every downstream regression, not a cosmetic adjustment. The full-sample shocks correlate 0.81 to 0.93 with a no-dummy baseline, confirming the sensitivity is real but not destabilising.

4.4 Diagnostics

The optimal model fits well and behaves like a clean structural system. The equation R^2 values are 0.43 for production growth, 0.92 for real activity, and 0.98 for the real oil price. The structural shocks are orthogonal to machine precision (largest off-diagonal correlation below 4×10^{-17}). Residuals show no autocorrelation (per-equation Ljung-Box and Durbin-Watson statistics near 2), and the oil-price residual is approximately Gaussian, although the production and activity residuals retain mild non-normality, as is usual for macro data. A simple sign check validates the economic labelling: the oil-specific shock correlates +0.76 with the contemporaneous oil-price change, the demand shock +0.08, and the supply shock essentially zero, exactly the pattern the identification predicts (supply moves the price only through multi-month VAR dynamics, not contemporaneously).

5 Stage Two: Sector Exposure via Local Projections

5.1 Estimator

With three clean shock series in hand, the second stage measures how each Nifty sleeve responds to each shock at horizons of zero to twelve months. The study uses local projections [6] rather than VAR-based impulse responses, because local projections are robust to misspecification of the dynamics and let every sector-horizon cell be estimated by a simple regression. For sector s, shock k, and horizon h,

$$\sum_{j=0}^{h} r_{s,\,t+j} \;=\; \alpha + \beta_{s,k,h}\,\text{shock}_{k,t} + \text{controls}_t + \varepsilon_{s,t+h},$$

where the dependent variable is the cumulative return over the next $h+1$ months. The controls are two lags of each shock, one lag of the sector's own return, and a COVID dummy. Because the cumulative dependent variable mechanically overlaps across horizons, inference uses Newey-West [12] heteroskedasticity-and-autocorrelation consistent standard errors with the bandwidth set to the overlap order. Crucially, betas are estimated on the training sample only (through February 2023), so no test-period return ever informs an exposure estimate.

5.2 The exposure map

Figure 4 traces the cumulative responses for four representative sleeves, and Figure 3 compresses the whole cross-section into a single quarterly snapshot. The economic story is coherent and survives the eye test. A positive oil-specific demand shock, the channel that maps most directly onto a rising oil price, lifts almost every equity sleeve, with the largest and most significant responses in Metal (+3.3% per one standard-deviation shock at three months), IT, Energy, and Bank. A supply-disruption shock does the opposite, pressing returns down, most visibly for IT and Metal. The aggregate demand shock is the cyclical signal: it is strongest for IT and the broad index, consistent with a growth-driven rally. The two safe-haven sleeves behave as hedges should, with Gold carrying a negative oil-specific loading and the liquid-cash fund essentially flat across all three channels.

5.3 Inference is taken seriously

Three independent inference methods are run side by side so that no single assumption carries the significance claims. The baseline HAC results are

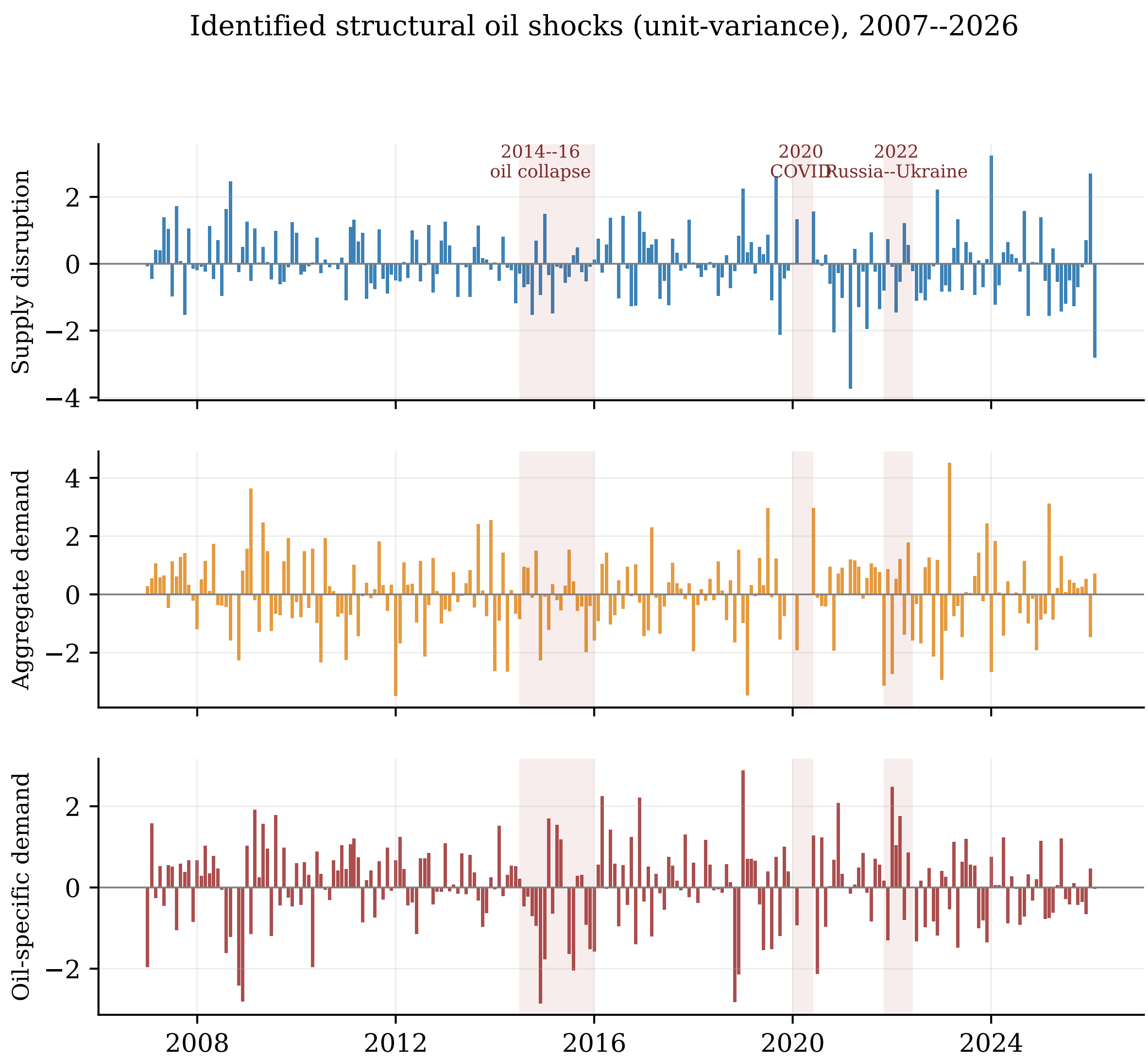

Figure 2: The three identified structural shocks (unit variance) over the equity sample. Shaded bands mark the three oil-stress episodes used throughout the study. The oil-specific series spikes sharply around the 2014–16 collapse and the 2022 Russia–Ukraine episode, while 2020 is dominated by demand and dummy-absorbed supply distortions.

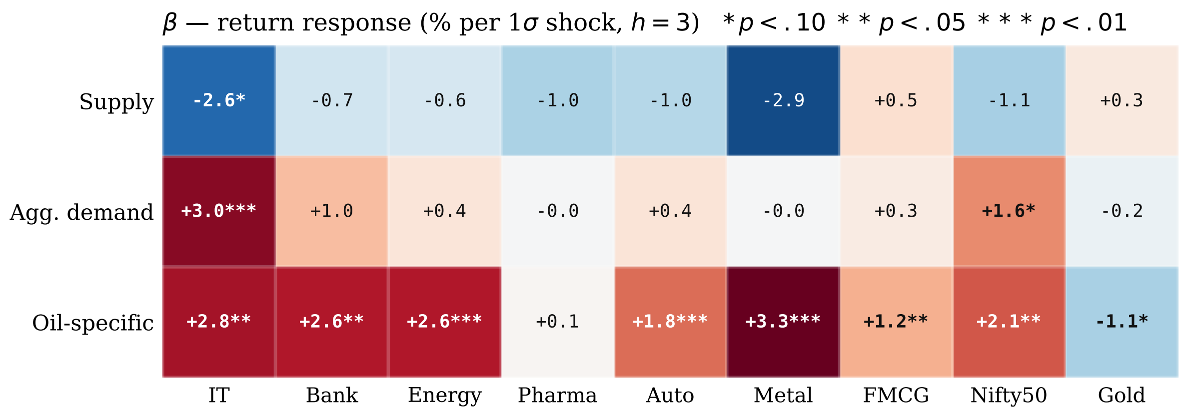

Figure 3: Quarterly ($h = 3$) return response of each sleeve to a one standard-deviation structural shock, in percent. Red is positive, blue negative; stars denote HAC significance ($*p<.10$, $**\ p<.05$, $***p<.01$). The oil-specific row is uniformly positive and the most heavily significant, which is what makes it the workhorse of the overlay.

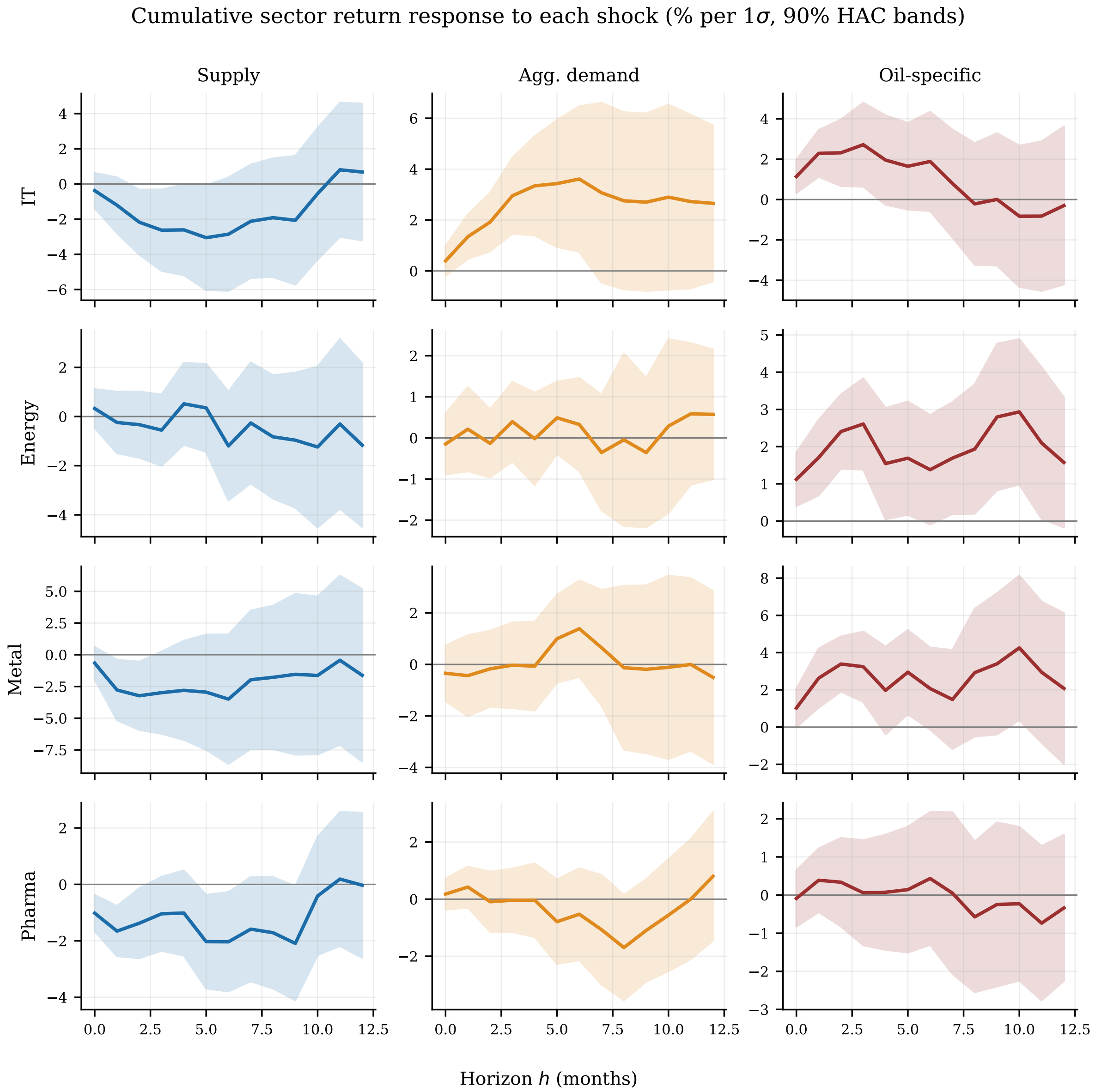

Figure 4: Cumulative local-projection impulse responses for four sleeves (rows) to each shock (columns), in percent per one standard-deviation shock, with 90% HAC bands. The oil-specific column peaks near the one-to-three month horizon, which motivates the choice of quarterly horizon for risk sizing.

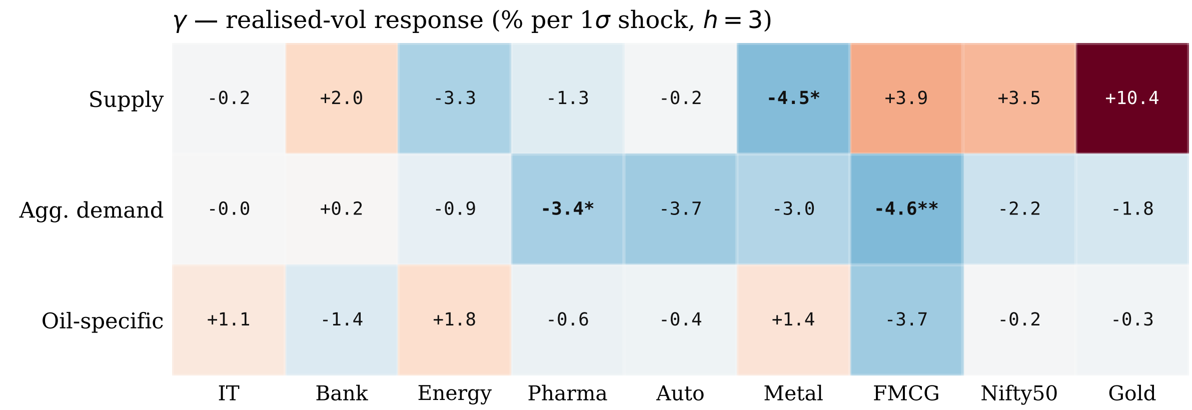

Figure 5: Quarterly volatility response ("gamma") of each sleeve to each shock. Almost no cell is significant, the visual signature of a null result. The overlay ignores this channel by design.

cross-checked against the lag-augmented approach of Montiel Olea and Plagborg-Møller [11], which widens standard errors by about 5% and overturns only a handful of marginal cells. A wild bootstrap in the style of Gonçalves and Kilian [4], with 195,000 draws, produces 90% intervals centred on the point estimates; the strongest oil-specific and supply cells survive at 90%, while three borderline cells are honestly demoted. The agreement across methods is the reason the exposure map can be trusted as an input to a trading rule.

5.4 A genuine null result on volatility

The same machinery is applied with log realised volatility as the dependent variable, the log transform being used because realised volatility is approximately Gaussian in logs [1]. The resulting "gamma" coefficients would capture an oil-driven *risk* channel. The result is a clean null: only 4% of the 150 gamma cells are significant at 5%, no more than chance would deliver (Figure 5). Oil shocks move the *level* of Indian sector returns, not their conditional volatility. This finding is allowed to shape the design: the overlay is built on return betas alone, and the gamma estimates are deliberately *not* used to generate signals. Reporting a null and acting on it is itself a piece of risk discipline.

6 Stage Three: Portfolio Construction

6.1 From betas to a signal

The exposure map is collapsed into a single effective beta per sector and shock by blending the one-, three-, and six-month horizons with precision weights, so that statistically sharper horizons count for more:

$$\hat{\beta}_{s,k} = \frac{\sum_h \beta_{s,k,h} / \hat{\sigma}_{s,k,h}}{\sum_h 1/\hat{\sigma}_{s,k,h}}, \qquad \hat{\sigma}_{s,k,h} = |\beta_{s,k,h}|/|t_{s,k,h}|.$$

The live shock environment is summarised by an exponentially weighted state $z_t = \rho\, z_{t-1} + \text{shock}_t$ with decay $\rho = 2/3$, which keeps a shock "alive" for about three months after it lands, matching the horizon at which the betas peak. Figure 6 shows the mechanism: an isolated monthly shock that would otherwise vanish becomes a persistent state variable the portfolio can actually trade on. The raw signal for each sector is then its beta vector dotted into a compressed shock state,

$$\text{sig}_{s,t} = \hat{\mathbf{B}}_s \cdot \tanh\big(z_t/c\big),$$

where the hyperbolic tangent with $c = 1$ caps the influence of extreme shocks, and the resulting vector is standardised across sectors before use.

6.2 Three portfolios run in parallel

Three books are run side by side and rebalanced monthly so that each marginal piece of machinery can be attributed. Portfolio A is the benchmark: an equal weight across the eight equity sleeves, scaled to a 15% annualised volatility target. Portfolio B-Lin is a transparent linear control that adds a signal tilt proportional to the standardised signal divided by each sector's downside volatility. Portfolio B-BL, the headline strategy, embeds the same oil views inside a Black-Litterman posterior [3],

$$\mu^{BL} = \big[(\tau\Sigma)^{-1} + \Omega^{-1}\big]^{-1}\big[(\tau\Sigma)^{-1}\Pi + \Omega^{-1}q\big],$$

where $\Pi = \delta\Sigma w^A$ is the equilibrium-implied return, the views q are scaled oil signals, and the view-confidence matrix Ω^{-1} is set from the statistical reliability of the underlying local-projection

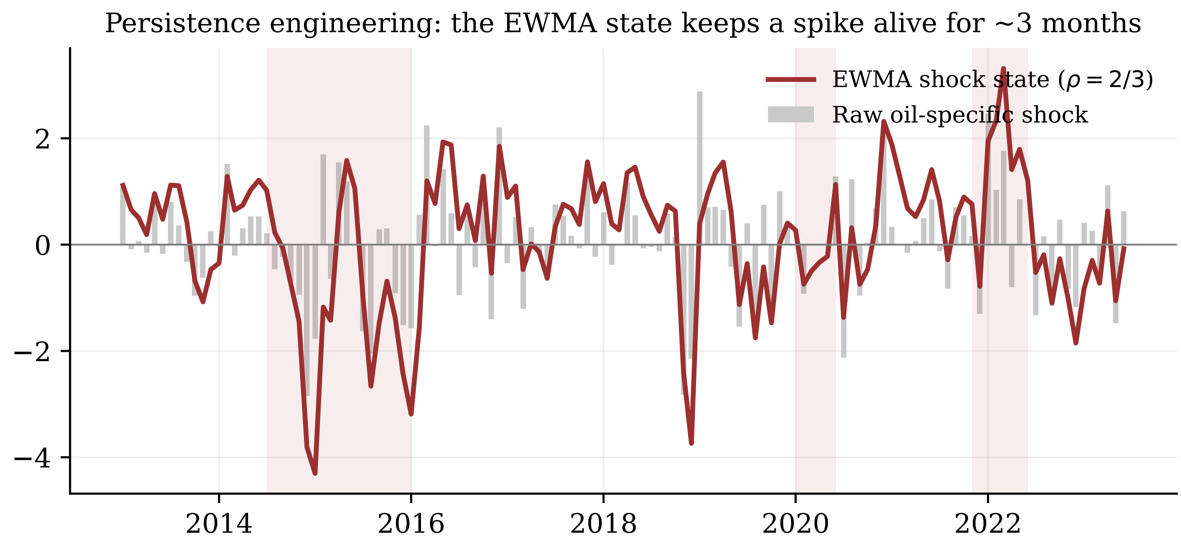

Figure 6: Raw oil-specific shock (grey bars) and its exponentially weighted state (red line). The state smooths and prolongs each spike so that the position stays elevated through the horizon at which the betas are largest.

estimates. This last point is the elegant part: a noisy beta automatically enters the portfolio as a low-confidence view and is shrunk toward equilibrium, so estimation uncertainty is handled inside the optimiser rather than ignored.

6.3 The conditional hedge sleeve

On top of the equity tilt, Portfolio B-BL carries a conditional hedge. When the normalised shock intensity exceeds one standard deviation, the strategy moves up to 15% of the book out of equities and into a gold and liquid-cash sleeve, split in proportion to each asset's defensive oil beta, with a 20% floor on each leg and a 20% hard cap on gold. The hedge is genuinely state-contingent: in calm months it is essentially absent, and it scales up smoothly as intensity rises. Finally, all three books are volatility-targeted to 15% using the trailing sector covariance, and a symmetric 5 basis-point half-spread is charged on turnover at every rebalance. The full parameter set is fixed before any out-of-sample result is examined, so the backtest is not the product of a tuning search.

7 Results

The backtest runs from August 2014, after a 36-month warm-up, to February 2026, splitting into 103 in-sample and 36 out-of-sample months. Table 3 reports the headline performance. In sample, the linear overlay does the most for raw Sharpe because it tilts aggressively, while the Black-Litterman book is the most conservative and the lowest in volatility. Out of sample, the ranking inverts in the strategy's favour: the Black-Litterman overlay lifts the net Sharpe ratio from the benchmark's 1.27 to 1.85, raises the annual net return from 11.9% to 16.2%, and does so at *lower* volatility (8.7% against 9.4%). The linear control, by contrast, underperforms out of sample, which is informative: the value is not in tilting hard but in tilting with the uncertainty-aware shrinkage that Black-Litterman provides.

Figure 7 tells the same story visually. The Black-Litterman book tracks the benchmark closely in quiet periods, by design it should not stray far when there is no oil signal, but pulls ahead through the stress windows and compounds the lead. The lower panel shows that its rolling volatility sits at or below the 15% target almost everywhere, and notably compresses during the 2020 and 2022 episodes where the benchmark's volatility spikes. The rolling Sharpe in Figure 8 confirms the out-of-sample separation is not a single lucky month.

7.1 The structural claim: less oil risk

The cleanest result is not about return at all. Decomposing each portfolio's variance into an oil-driven part and the rest, the overlay cuts the oil-variance share of the book from 1.75% to 0.84% out of sample, a reduction of about 0.9 percentage points, or roughly half. Table 4 collects the attribution. Two facts stand out. First, the Black-Litterman book achieves its risk reduction with far lower turnover than the linear control (0.83 versus 14.3 annualised), because most of the work is done by a small, well-targeted hedge rather than by constant re-tilting. Second, its gold allocation is

Table 3: Net performance, in-sample (through 2023-02) and out-of-sample (from 2023-03). Returns and volatilities are annualised; costs are 5 bps per side.

Period	Strategy	Ret %	Vol %	Sharpe	MaxDD %
In-sample	Base: equal weight	5.79	12.08	0.479	−29.86
	Linear overlay	8.32	11.03	0.754	−14.65
	BL overlay	6.04	10.37	0.583	−23.99
Out-of-sample	Base: equal weight	11.86	9.37	1.266	−15.02
	Linear overlay	9.94	10.79	0.921	−10.81
	BL overlay	**16.19**	**8.74**	**1.853**	−13.38

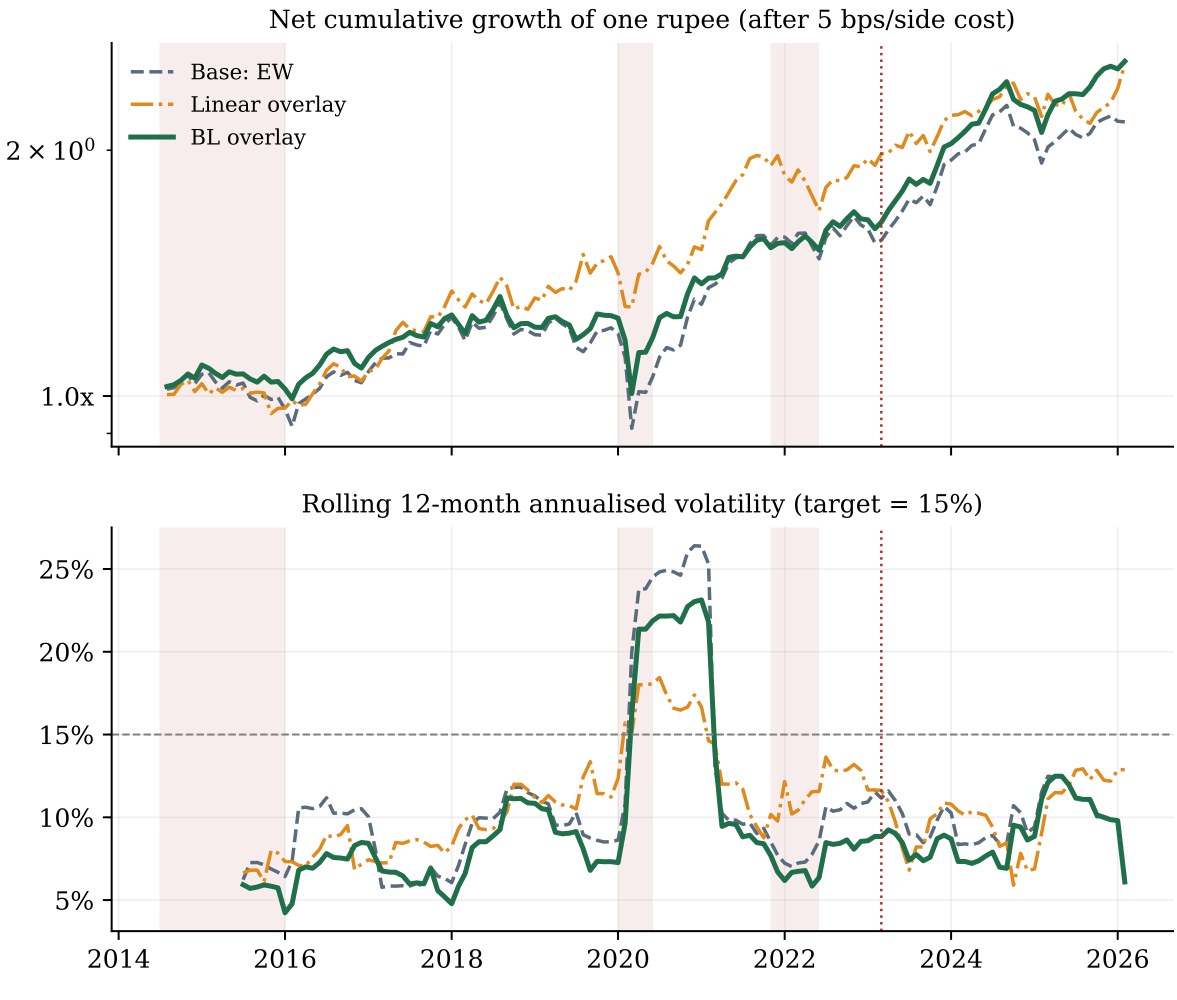

Figure 7: Top: net cumulative growth of one rupee (log scale, 5 bps per side). Bottom: rolling twelve-month annualised volatility against the 15% target. Shading marks oil episodes; the dotted line marks the start of the out-of-sample period.

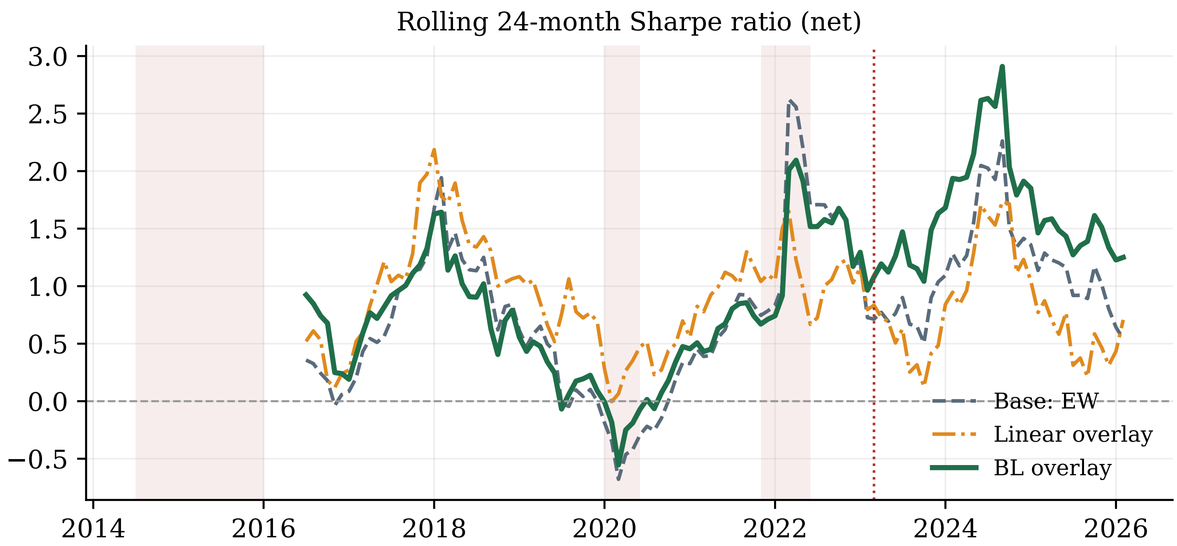

Figure 8: Rolling 24-month net Sharpe ratio. The Black-Litterman overlay holds a persistent edge through the out-of-sample window rather than in a single spike.

strongly correlated with oil-shock intensity (0.68), which is the signature of a hedge that is actually conditional rather than a fixed bet on gold.

Table 4: Attribution of the two overlays against the benchmark (out-of-sample unless noted).

	Base	Linear	**BL**
Sharpe, in-sample	0.479	0.754	0.583
Sharpe, out-of-sample	1.266	0.921	**1.853**
Oil-variance share (%)	1.75	1.32	**0.84**
Oil-share reduction (pp)	–	0.42	**0.91**
Tracking error (%)	–	9.59	2.85
Turnover (annualised)	–	14.30	**0.83**
Avg. gold weight (%)	0.0	10.5	12.7
corr(gold, intensity)	–	0.04	**0.68**

Figure 9 places the three books against the long-only efficient frontier estimated from training data. Out of sample the Black-Litterman portfolio sits up and to the left of the benchmark, the direction every risk manager wants, and indeed pokes through the in-sample frontier because the realised out-of-sample period rewarded its defensive tilt. Figure 10 shows where the weight actually goes year by year, with the gold and liquid-cash rows lighting up precisely in the oil-stress years.

8 Robustness

A strategy is only as credible as the attempts made to break it. This section reports those attempts, including the ones that the strategy fails. Throughout, inference leans on autocorrelation-robust standard errors and resampling rather than on a single test statistic, and reported Sharpe ratios are read against the deflated Sharpe benchmark of Bailey and López de Prado [2], which adjusts for the number of specifications a backtest implicitly searches.

8.1 Walk-forward folds

The overlay is re-run across four walk-forward folds, each centred on a different oil episode, with betas frozen at the fold's training cutoff and no per-fold re-tuning. Figure 11 shows the result: the overlay's out-of-sample Sharpe exceeds the benchmark in every fold, most dramatically in the 2014–16 collapse, where it turns a negative benchmark Sharpe (-0.29) positive ($+0.37$). In every episode-containing fold the overlay also reduces the maximum drawdown, by 2.6 to 3.8 percentage points.

8.2 Where the risk reduction comes from

Two extensions confirm that the oil-specific channel does the heavy lifting. Figure 12 decomposes the oil-variance reduction by channel: the oil-specific shock accounts for the largest single share (about 44%), with supply contributing 39% and aggregate demand the rest. A bootstrap pins the oil-specific fraction tightly at 43.7% with a 95% interval of $[43.1, 44.1]$%. An ablation that zeroes the supply and demand betas and keeps only the oil-specific channel retains 97 to 100% of the episode protection, so the oil-specific signal essentially carries the hedge on its own.

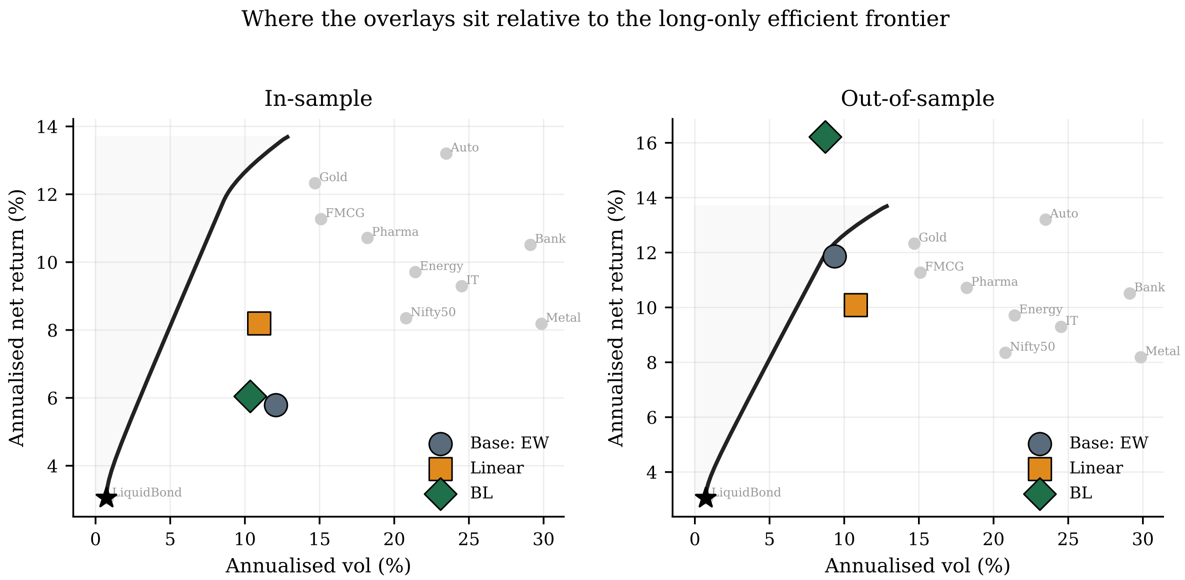

Figure 9: The three portfolios against the long-only mean-variance frontier (train-period estimates). Grey dots are individual sleeves; the star is the minimum-variance portfolio. Out of sample the BL overlay sits northwest of the benchmark.

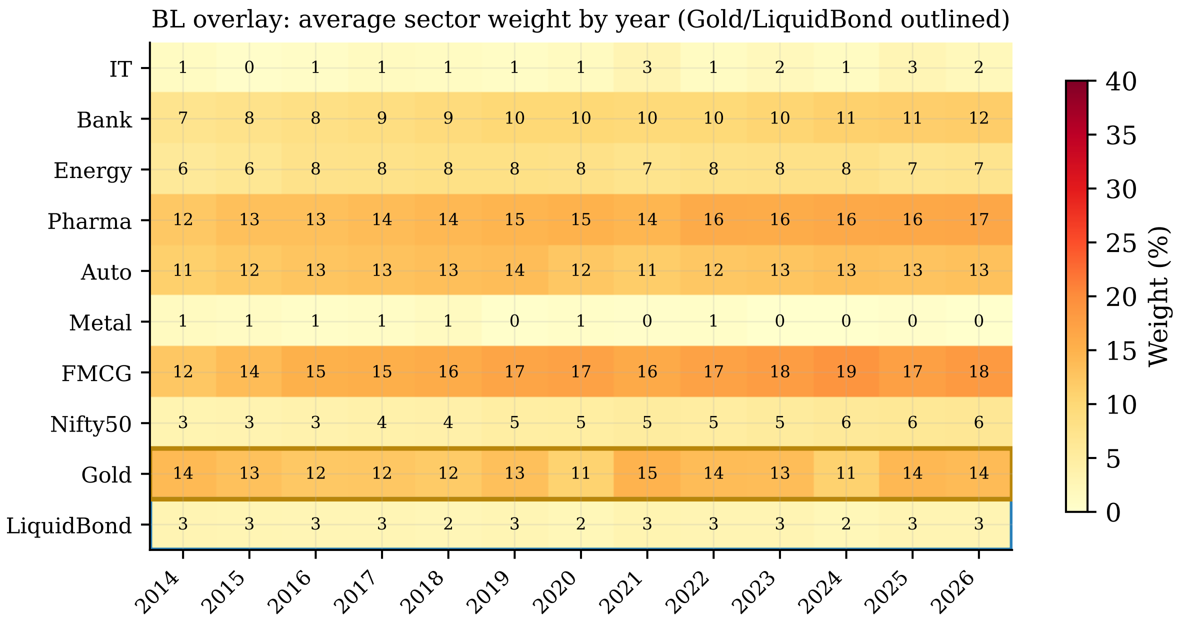

Figure 10: Average BL-overlay weight by sleeve and year. The gold and liquid-bond rows (outlined) carry weight only in oil-stress years, confirming the hedge is conditional.

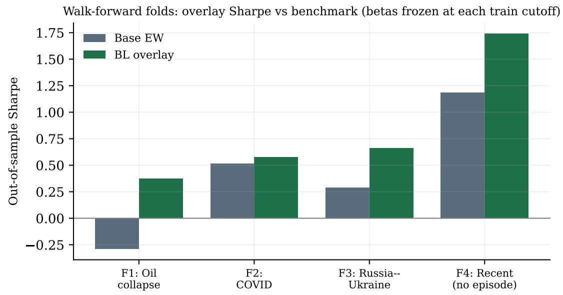

Figure 11: Out-of-sample Sharpe by walk-forward fold, benchmark versus BL overlay. The overlay improves every fold, including the one (2014–16) where the benchmark loses money.

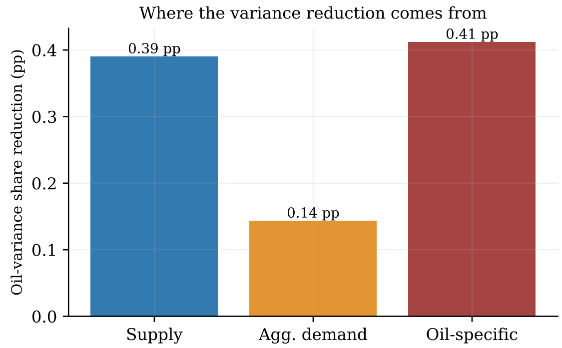

Figure 12: Oil-variance-share reduction attributed to each shock channel. The oil-specific channel is the largest single contributor.

8.3 Do the betas generalise, or just memorise?

Two placebo-style tests probe whether the betas carry real information. In a leave-one-episode-out exercise, the betas are re-estimated with each target episode (plus a buffer) excised, and still deliver 91 to 97% of their protection on the non-COVID episodes, so the exposure map is not merely memorising history. A permutation placebo that shuffles the sector betas at random achieves a median episode protection of only 1.62 percentage points against the real overlay's 2.41, with a pseudo p-value of 0.08, so the real betas beat random assignment. Across temporal folds the sign of the betas agrees with the full-sample sign 82% of the time, and 88% for the all-important oil-specific channel.

8.4 Conditionality and cost

The gold sleeve is conditional in the strict sense (Figure 13): its weight correlates +0.68 with oil-shock intensity, averages 14.4% in the top third of intensity against 9.6% in the bottom third, and the 20% gold cap binds in zero months, so the cap shapes only episode peaks. A parameter-sensitivity grid over the hedge budget and split floor leaves both the episode protection and the oil-share reduction positive in all nine cells, with no sign reversals.

The strategy also survives cost: stepping the transaction charge from 0 to 20 basis points per side leaves the out-of-sample Sharpe advantage essentially intact (the edge falls only from 0.589 to 0.580), because the Black-Litterman book trades so little.

8.5 The honest negatives

Two tests do not go the strategy's way, and both are reported in full because they bound what can legitimately be claimed.

Timing versus a static tilt. A static defensive benchmark, a constant tilt of the same average size as the dynamic overlay, is built and compared. Over the full sample the static tilt actually delivers *more* episode protection than the dynamic timing (2.92 against 2.41 percentage points), and a moving-block bootstrap [9] puts the difference $D = \text{static} - \text{dynamic}$ at +0.74 pp with a 95% interval of $[0.23, 1.56]$ that excludes zero. Out of sample the interval $[-0.22, 0.79]$ brackets zero. The honest conclusion is that, with only about three independent oil episodes in the sample, the value of *timing* the hedge cannot be statistically distinguished from simply holding a constant defensive posture. The structural variance reduction is real; the timing alpha is unproven.

A placebo that works too well. When the oil-specific shock that drives the hedge is swapped for the supply shock, the placebo retains 99% of the protection. Read strictly, this means the protection may owe as much to a general "risk-off" tilt toward gold during turbulent macro months as to anything uniquely oil-specific. This is reported rather than rationalised away.

9 Limitations

Beyond the two negative tests, the study is bounded by four structural caveats. The equity data carry survivorship bias, so return *levels* are upward biased and the analysis leans deliberately on risk and relative-stress metrics. Statistical power is genuinely low: monthly data with a handful of distinct oil regimes produce wide Sharpe-difference intervals, and the spanning-style tests should not be over-read. The betas are point estimates fixed from the training window and may drift in real time, although the Black-Litterman view variance absorbs some of that uncertainty. Finally, sector-level betas are applied uniformly to all constituents; a second layer that distributes the sector signal across individual stocks is architecturally supported but not implemented here.

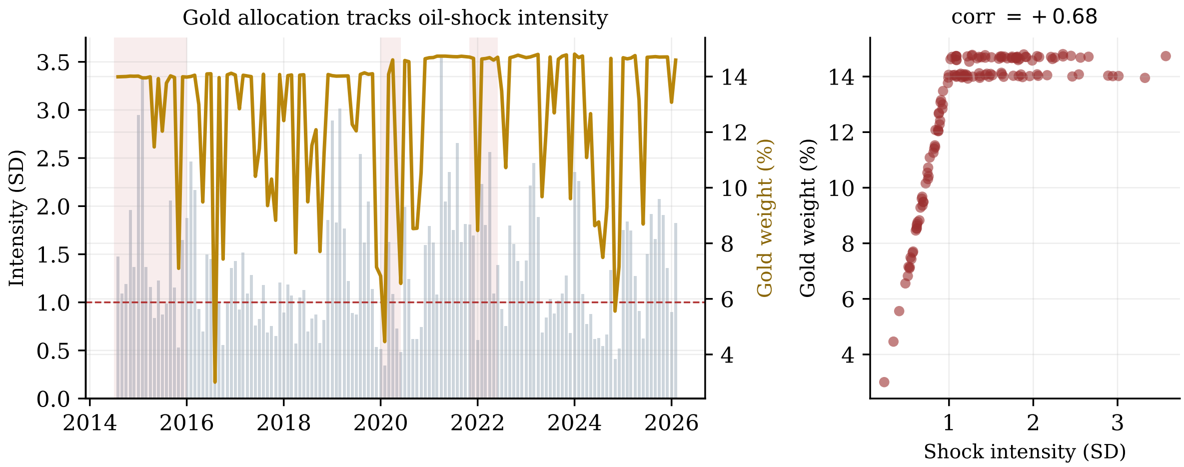

Figure 13: The gold allocation is genuinely state-contingent. Left: gold weight (gold line) rises and falls with oil-shock intensity (grey bars). Right: the two line up with a correlation of +0.68.

10 Conclusion

This study set out to test whether the Kilian [7] decomposition of oil shocks can be turned into a usable risk-management overlay for Indian equities, and it returns a measured verdict. The decomposition is sound: the three shocks are cleanly identified, economically sensible, and stable across subsamples. The exposure map they imply is statistically robust under three separate inference regimes and is dominated by an oil-specific demand channel that lifts cyclical and energy-linked sectors while gold leans the other way. Built on that map, a Black-Litterman overlay with a conditional gold hedge roughly halves the portfolio's oil-variance share and, out of sample, improves the net Sharpe ratio from 1.27 to 1.85 at lower volatility, with the gains concentrated exactly where they should be, in oil-stress windows, and at negligible turnover.

The contribution is as much methodological as empirical. By running a transparent linear control alongside the headline strategy, by reporting a genuine null on the volatility channel and acting on it, and above all by showing where the evidence runs out, that hedge *timing* cannot be separated from a static tilt and that a supply-shock placebo protects nearly as well, the study models the kind of disciplined, self-skeptical risk research that the structural variance-reduction result, the part that survives every test, deserves.

A Full Exposure Estimates

Table 5 reports the quarterly ($h = 3$) cumulative return response of every sleeve to each shock, the snapshot used for risk sizing. Table 6 reports the multi-horizon effective betas actually fed to the overlay, blended across the one-, three-, and six-month horizons with precision weights. Both are estimated on training data only (through February 2023).

Table 5: Quarterly ($h = 3$) return response, percent per one standard-deviation shock. HAC significance: $*p<.10$, $**\ p<.05$, $***p<.01$.

Sleeve	Supply	Agg. demand	Oil-specific
Auto	−0.82	+0.52	+1.85 ***
Bank	−0.60	+1.10	+2.62 **
Energy	−0.49	+0.44	+2.61 ***
FMCG	+0.52	+0.28	+1.16 **
Gold	+0.34	−0.24	−1.07 *
IT	−2.56 *	+3.00 ***	+2.71 **
LiquidBond	−0.06	−0.02	−0.03
Metal	−2.83	+0.09	+3.26 ***
Nifty 50	−1.06	+1.60 *	+2.04 **
Pharma	−1.01	−0.06	+0.07

Table 6: Multi-horizon effective betas (precision-weighted over $h \in \{1, 3, 6\}$), in raw log-return units, as used by the overlay signal.

Sleeve	Supply	Agg. demand	Oil-specific
IT	−0.0225	+0.0264	+0.0235
Bank	−0.0088	+0.0125	+0.0196
Energy	−0.0075	+0.0038	+0.0205
Pharma	−0.0162	+0.0001	+0.0039
Auto	−0.0069	+0.0041	+0.0152
Metal	−0.0286	+0.0108	+0.0276
FMCG	+0.0046	−0.0006	+0.0111
Nifty 50	−0.0119	+0.0131	+0.0163
Gold	+0.0077	+0.0013	−0.0076
LiquidBond	−0.0005	−0.0002	−0.0003

Table 7: Key configuration parameters.

Parameter	Value	Meaning
Vol. target	0.15	Annualised portfolio volatility target
Max sector weight	0.25	Cap on any single sector
λ (overlay)	0.05	Overlay strength (grid-searched on train)
Transaction cost	5 bps	One-way half-spread on turnover
Signal lag	1 mo	Data-availability lag
Warm-up	36 mo	History before first rebalance
Shock threshold	1.0 SD	Intensity that activates the hedge
EWMA decay ρ	0.667	Shock-state persistence $(1 - 1/H)$
Hedge max	0.15	Maximum hedge-sleeve allocation
Gold cap	0.20	Hard cap on gold weight
BL risk aversion δ	2.5	Black-Litterman equilibrium parameter
BL uncertainty τ	0.05	Scaling on equilibrium returns
Bootstrap block	6 mo	Moving-block length
Bootstrap draws	5,000	Replications per test

B Strategy Configuration

Table 7 lists the key hyperparameters. Every value was pinned before the out-of-sample period was examined; only the overlay strength λ was selected, once, by a grid search restricted to the training window.

C Robustness Scorecard

Table 8 collects the verdict of every robustness test in one place, including the two the strategy fails.

Table 8: Summary of robustness tests and verdicts.

Test	Result	Verdict
Oil-share reduction (bootstrap)	0.93 pp, CI $[0.91, 0.96]$	Pass
Walk-forward folds	Sharpe up in 4/4	Pass
Channel decomposition	oil-specific 43.7%	Pass
Signal ablation	97–100% retained	Pass
Leave-one-episode-out	91–97% retained	Pass
Permutation placebo	pseudo-$p = 0.08$	Pass
Beta sign stability	82% across folds	Pass
Gold conditionality	corr +0.68	Pass
Parameter grid	positive 9/9 cells	Pass
Transaction-cost grid	edge intact to 20 bps	Pass
Timing vs static tilt	static $\geq$ dynamic	*Fail*
Supply-shock placebo	99% retained	*Fail*

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